

AHMED GENEED

Electronics & Communication Engineer | Software, Agentic AI & Audio DSP Architect

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EXECUTIVE SUMMARY

High-performance Electronics & Communication Engineer specializing in full-stack software architecture, cryptographic identity frameworks, and high-precision audio digital signal processing (DSP). Proven expertise in orchestrating agentic AI systems, Model Context Protocol (MCP) integrations, multi-model fallback chains, and real-time computer vision hardware. Author of multiple open-source software libraries built for deterministic execution across local edge devices and cloud environments.

TECHNICAL SKILLS & COMPETENCIES

Agentic AI & Prompt Eng:	Hermes Agent, OpenClaw, Autonomous Subagents, System Prompting, MCP, Claude, Google Gemini, DeepSeek, OpenAI GPT
Multimodal Generative AI:	Audio/Music Generation (Suno, Mureka, HeartMuLa, Whisper), Video Synthesis (SeaDance, MuseSpark, Wan, Kling, FLUX)
Languages & Web UI:	Python, C++, C#, C, MySQL, MATLAB, Embedded C, OpenCV, Gradio Web Interfaces
Audio DSP & Signal Eng:	Multiband Mid/Side Phase Correlation, Bandwidth Extension (BWE), ITU-R BS.1770-4 LUFS, True Peak Limiting, Pedalboard, SciPy, NumPy
Hardware & Security Systems:	OpenCV Facial Recognition, RFID Access Systems, Arduino Mega Microcontrollers, SHA-512 Biometrics, Zero-Trust Architecture, CCNA
Knowledge & Memory Vaults:	Obsidian Knowledge Graphs, Persistent AI Memory Architectures, Vector Search, Local SQLite FTS5

FLAGSHIP OPEN-SOURCE PROJECTS

HORUS KEY — Sovereign FinTech & Cryptographic Identity Monolith | github.com/G3need/HORUS

- Architected a zero-trust travel identity platform using SHA-512 biometric facial hashing and arrival compliance rules.
- Integrated automated multi-model AI fallback reasoning chains (Gemini Flash) with Gradio web interfaces and Colab launchers.

Sovereign-DSP — Autonomous Audio Mastering & Phase-Linear Signal Processing Suite |

github.com/G3need/Sovereign-DSP

- Engineered a Python DSP pipeline featuring Multiband Mid/Side phase correlation locks, harmonic Bandwidth Extension (BWE), and ITU-R BS.1770-4 LUFS compliance.
- Built interactive Google Colab execution scripts with parameterized form controls for drag-and-drop batch processing.

FOOTY4ALL — RFID & Computer Vision Stadium Security System | github.com/G3need/FOOTY4ALL

- Developed an award-winning (3rd Place Egyptian Engineering Day 2016) hardware/software stadium access control ecosystem.
- Implemented real-time C++ OpenCV facial recognition, Qt5 management GUIs, and Arduino Mega solenoid gate controllers.

PROFESSIONAL EXPERIENCE

Telecom Egypt (WE) — Outlets & Visits Technical Support Specialist | Ismailia, Egypt

06/2021 – Present

- Remotely diagnose and resolve complex connectivity, routing, and digital service infrastructure incidents via enterprise Ticketing Systems.
- Oversee hardware outlet inventory stocks and compile daily operational performance telemetry reports for corporate headquarters.

Egyptian Air Force — Personnel Affairs Office Manager | Egypt

12/2017 – 12/2020

- Supervised full operation center personnel records, logistics management, and database synchronization across secure internal networks.
- Automated document tracking, incoming/outgoing official communications logs, and encrypted digital records filing procedures.

Yellow Works Ltd. — Co-Founder & Lead Systems Engineer | Ismailia, Egypt

2015 – 2017

- Co-founded technology enterprise to execute the Ismailia Stadium physical security pilot project incorporating RFID and computer vision.

EDUCATION & AWARDS

Misr International University (MIU) — B.Sc. Electronics & Communication Engineering (2017) | Major GPA: 3.76

- Egyptian Engineering Day (EED 2016):** 3rd Place Recipient for FOOTY4ALL Stadium Access Security System.
- MIU Graduation Project Achievement Award:** Recognized for outstanding engineering innovation.

American International School — High School Diploma, Dubai, UAE (2011) | GPA: 3.8